

PAPER_1398 — UQFF Resolution of the Sleeping Beauty Problem (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Anthropic Probability (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "sleeping_beauty"})` **Calculator surface:** `calculate_paradox({"paradox": "sleeping_beauty"})` **Closure helper:** `_l96_uqff_axiom_sleeping_beauty_closure()`

The Paradox

SB protocol: coin flipped Sunday. Heads → wake once Monday. Tails → wake Monday AND Tuesday with memory erased between. Upon awakening, what credence to assign to heads? Halfer says 1/2, thirder says 1/3.

UQFF Closed Identity

$P(\text{heads} \mid \text{awake}) = 1/(D_{\text{phys}} - 1) = 1/3$ EXACT (thirder position)

$F_U = 1$ global ledger normalization across the awakening states. Three F_U -equally-weighted awakenings: (heads, Monday), (tails, Monday), (tails, Tuesday). $P(\text{heads} \mid \text{awake}) = 1/3 = 1/(D_{\text{phys}} - 1)$ EXACT.

Physical Interpretation

UQFF derives the thirder position from $F_U = 1$ normalization. The halfer position would require treating epistemic-state-counting differently from ledger-state-counting; UQFF asserts they are the same.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "sleeping_beauty"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_sleeping_beauty_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_sleeping_beauty_closure`
- Dispatch keys: `sleeping_beauty`, `sleeping_beauty_problem`
- Related foundational papers: PAPER_646 ($F_U B_i / F_U=1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF dispatch closure for the Sleeping Beauty Problem via the identity above.